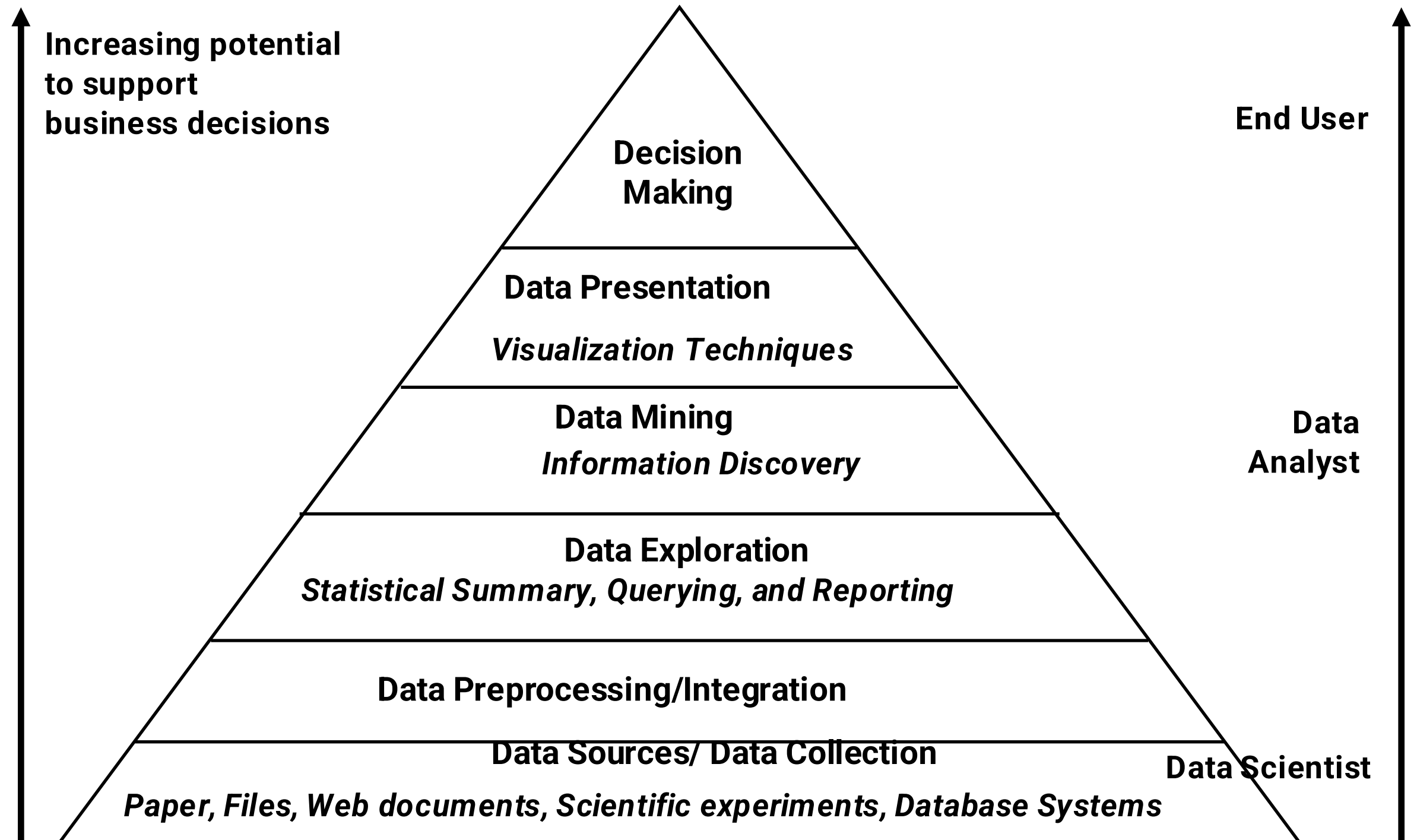


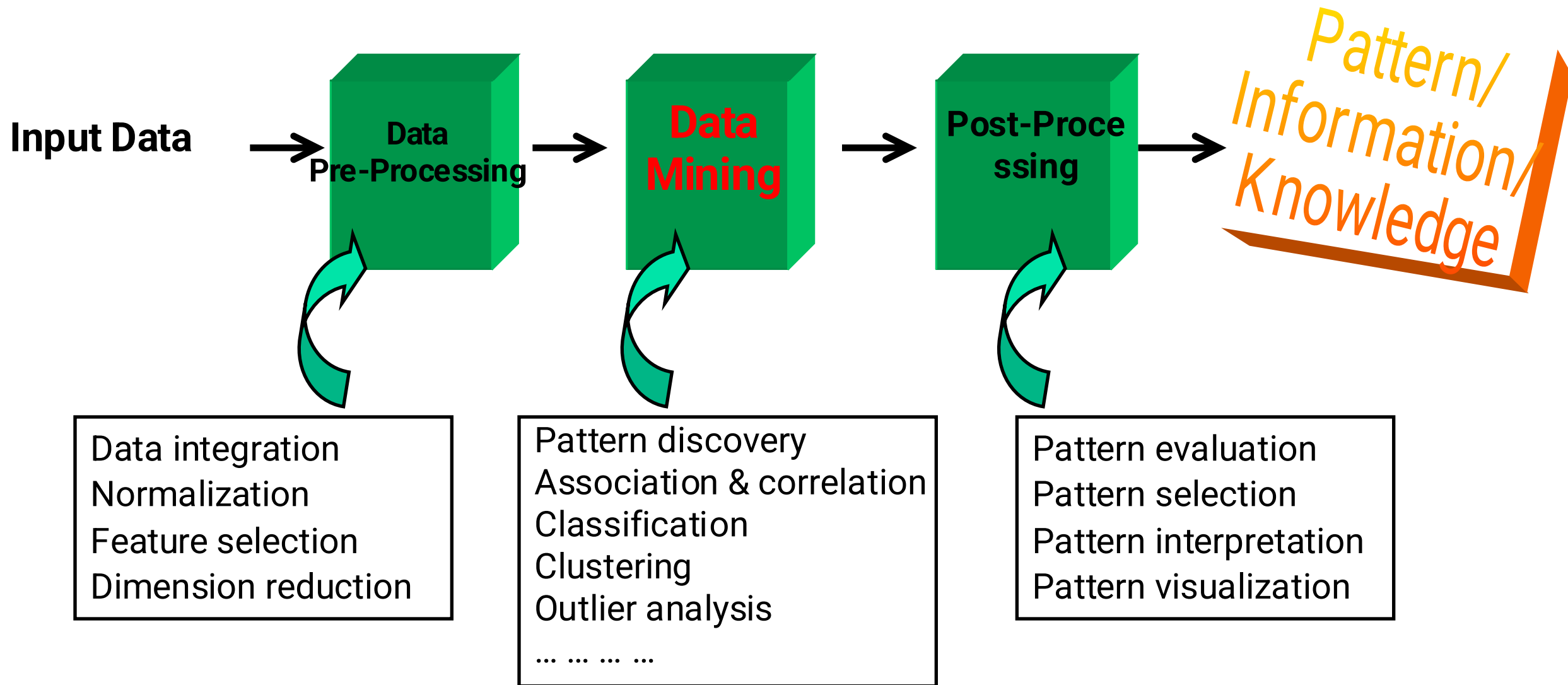
Data Science and Business Intelligence



Data Preprocessing

- Data preprocessing
 - **Data cleaning** (remove noise and inconsistent data)
 - **Data integration** (multiple data sources maybe combined)
 - **Data selection** (data relevant to the analysis task are retrieved from database)
 - **Data transformation** (data transformed or consolidated into forms appropriate for mining)
- Data Modeling and Analysis (Done with data preprocessing)
 - **Data mining** (an essential process where intelligent methods are applied to extract data patterns)
 - **Pattern evaluation** (indentify the truly interesting patterns)
 - **Knowledge presentation** (mined knowledge is presented to the user with visualization or representation techniques)
- Deployment (use of discovered knowledge)

A Typical View from ML and Statistics



- This is a view from typical machine learning and statistics communities

Multi-Dimensional View of Data Mining

■ Data to be mined

- Database data (extended-relational, object-oriented, heterogeneous, legacy), transactional data, stream, spatiotemporal, time-series, sequence, text and web, multi-media, graphs & social and information networks

■ Knowledge to be mined (or: Data mining functions)

- Regression, association, classification, clustering, trend/deviation, outlier analysis, summarization, etc.
- Descriptive vs. predictive data mining
- Multiple/integrated functions and mining at multiple levels

■ Techniques utilized

- Machine learning, statistics, pattern recognition, visualization, Natural Language Processing, Image Processing, etc.

■ Applications adapted

- Retail, telecommunication, banking, fraud analysis, bio-data mining, stock market analysis, text mining, Web mining, etc.

Different Kinds of Datasets



- Database-oriented data sets and applications
 - Relational database, data warehouse, transactional database
- Advanced data sets and advanced applications
 - Data streams and sensor data
 - Time-series data, temporal data, sequence data (incl. bio-sequences)
 - Structure data, graphs, social networks and multi-linked data
 - Object-relational databases
 - Spatial data and spatiotemporal data (like GPS tracks, weather patterns, or crime incidents, using data points with coordinates (latitude/longitude) and timestamps)
 - Multimedia database
 - Text databases
 - The World-Wide Web

Relational Databases

- DBMS – database management system, contains a collection of interrelated databases
e.g. Faculty database, student database, publications database
- Each database contains a collection of tables and functions to manage and access the data.
e.g. student_bio, student_graduation, student_parking
- Each table contains columns and rows, with columns as attributes of data and rows as records.
- Tables can be used to represent the relationships between or among multiple tables.

Relational Databases – *AllElectronics* store

customer

<u>cust_ID</u>	name	address	age	income	credit_info	category	...
C1	Smith, Sandy	1223 Lake Ave., Chicago, IL	31	\$78000	1	3	...
...	...	...	...	...	...	...	...

item

<u>item_ID</u>	name	brand	category	type	price	place_made	supplier	cost
I3	hi-res-TV	Toshiba	high resolution	TV	\$988.00	Japan	NikoX	\$600.00
I8	Laptop	Dell	laptop	computer	\$1369.00	USA	Dell	\$983.00
...	...	...	...	...	...	...	...	...

employee

<u>empl_ID</u>	name	category	group	salary	commission
E55	Jones, Jane	home entertainment	manager	\$118,000	2%
...	...	...	...	...	...

branch

<u>branch_ID</u>	name	address
B1	City Square	396 Michigan Ave., Chicago, IL
...	...	...

purchases

<u>trans_ID</u>	cust_ID	empl_ID	date	time	method_paid	amount
T100	C1	E55	03/21/2005	15:45	Visa	\$1357.00
...	...	...	...	...	...	...

items_sold

<u>trans_ID</u>	<u>item_ID</u>	qty
T100	I3	1
T100	I8	2
...	...	...

works_at

<u>empl_ID</u>	<u>branch_ID</u>
E55	B1
...	...

Relational Databases

- With a relational query language, e.g. SQL, we will be able to find answers to questions such as:
 - How many items were sold last year?
 - Who has earned commissions higher than 10%?
 - What is the total sales of last month for Dell laptops?
- When data mining is applied to relational databases, we can search for trends or data patterns.
- Relational databases are one of the most commonly available and rich information repositories, and thus are a major data form in our study.

Transactional Databases

- Consists of a file where each record represents a transaction
- A transaction typically includes a unique transaction ID and a list of the items making up the transaction.

<i>trans_ID</i>	<i>list of item_IDs</i>
T100	I1, I3, I8, I16
T200	I2, I8
...	...

- Either stored in a flat file or unfolded into relational tables
- Easy to identify items that are frequently sold together

Data Mining Tasks

- Prediction Tasks

- Use some variables to predict unknown or future values of other variables

- Description Tasks

- Find human-interpretable patterns that describe the data.

Common data mining tasks

- Classification [Predictive]
- Clustering [Descriptive]
- Association Rule Discovery [Descriptive]
- Sequential Pattern Discovery [Descriptive]
- Regression [Predictive]
- Deviation Detection [Predictive]

Classification: A Predictive task

- Given a collection of records (*training set*)
 - Each record contains a set of *attributes*, one of the attributes is the *class*.
- Find a *model* for class attribute as a function of the values of other attributes.
- Goal: previously unseen records should be assigned a class as accurately as possible.
 - A *test set* is used to determine the accuracy of the model. Usually, the given data set is divided into training and test sets, with training set used to build the model and test set used to validate it.

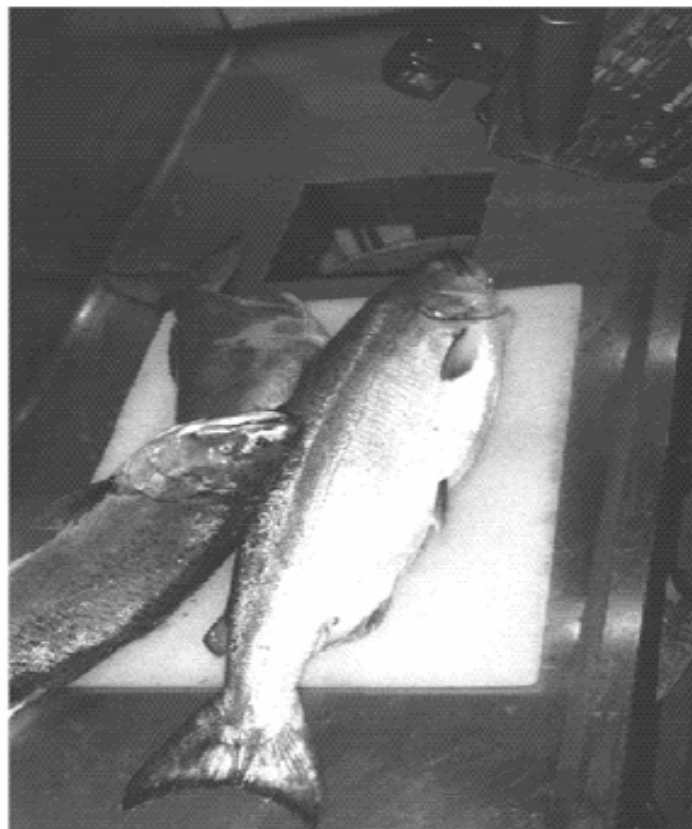
Classification: An Example

(from *Pattern Classification* by Duda & Hart & Stork – Second Edition, 2001)

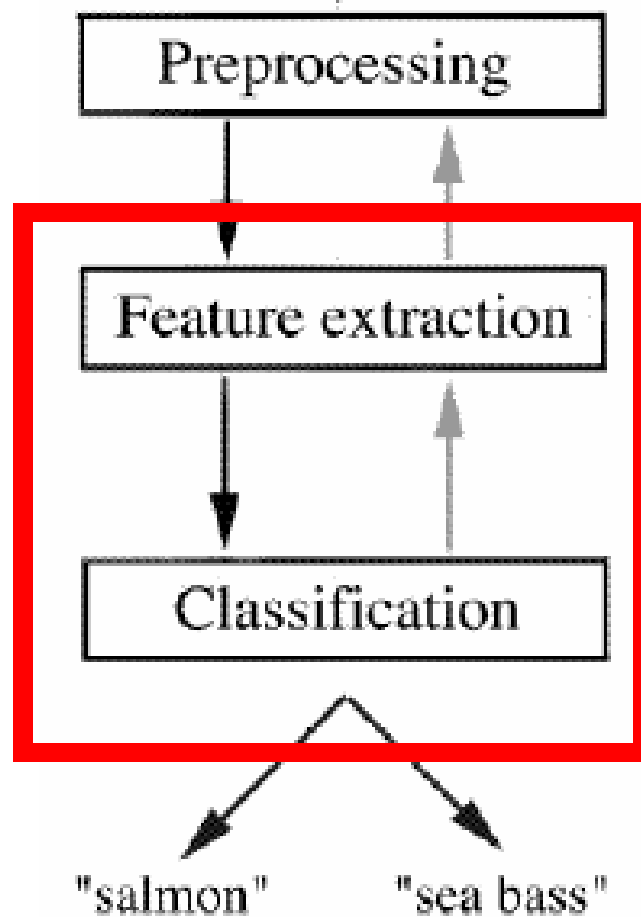
- A fish-packing plant wants to automate the process of sorting incoming fish according to species
- As a pilot project, it is decided to try to separate sea bass from salmon using optical sensing

Classification

- Features (to distinguish):
 - (i) Length
 - (ii) Lightness
 - (iii) Width
 - (iv) Position of mouth



Classification



- **Preprocessing:** Images of different fishes are isolated from one another and from background;
- **Feature extraction:** The information of a single fish is then sent to a feature extractor, that measure certain " features" or " properties" ;
- **Classification:** The values of these features are passed to a classifier that evaluates the evidence presented, and build a model to discriminate between the two species

Classification

- Domain knowledge:
 - A sea bass is generally longer than a salmon
- Related feature: (or attribute)
 - Length
- Training the classifier:
 - Some examples are provided to the classifier in this form: <fish_length, fish_name>
 - These examples are called training examples
 - The classifier *learns* itself from the training examples, how to distinguish Salmon from Bass based on the *fish_length*

Classification

n

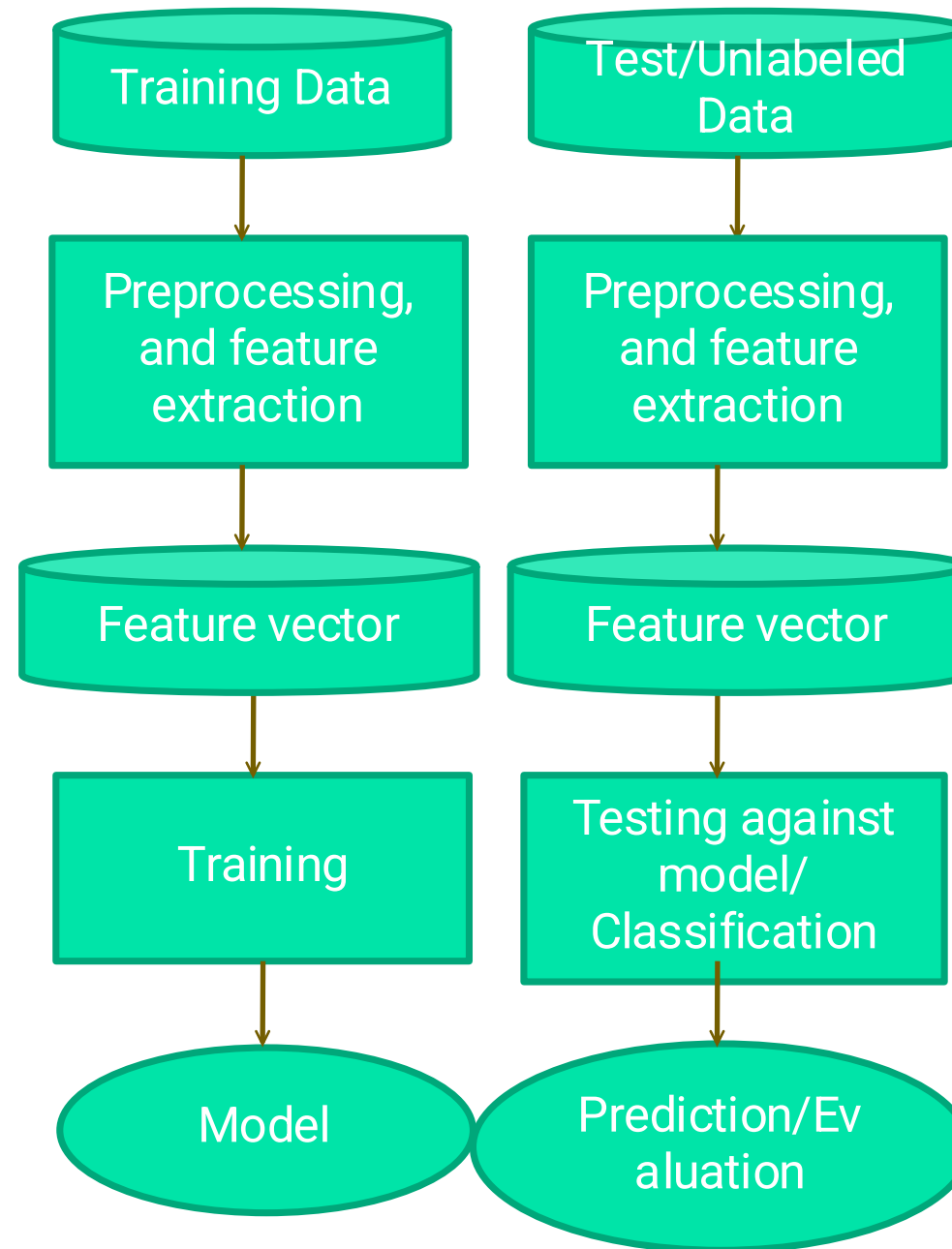
- Classification model (hypothesis):
 - The classifier generates a model from the training data to classify future examples (test examples)
 - An example of the model is a rule like:

Rule: If $Length \geq l^$ then sea bass otherwise salmon*
 - Here the value of l^* determined by the classifier
- Testing the model
 - Once we get a model out of the classifier, we may use the classifier to test future examples.
 - The test data is provided in the form $\langle fish_length \rangle$
 - The classifier outputs $\langle fish_type \rangle$ by checking *fish_length* against the model

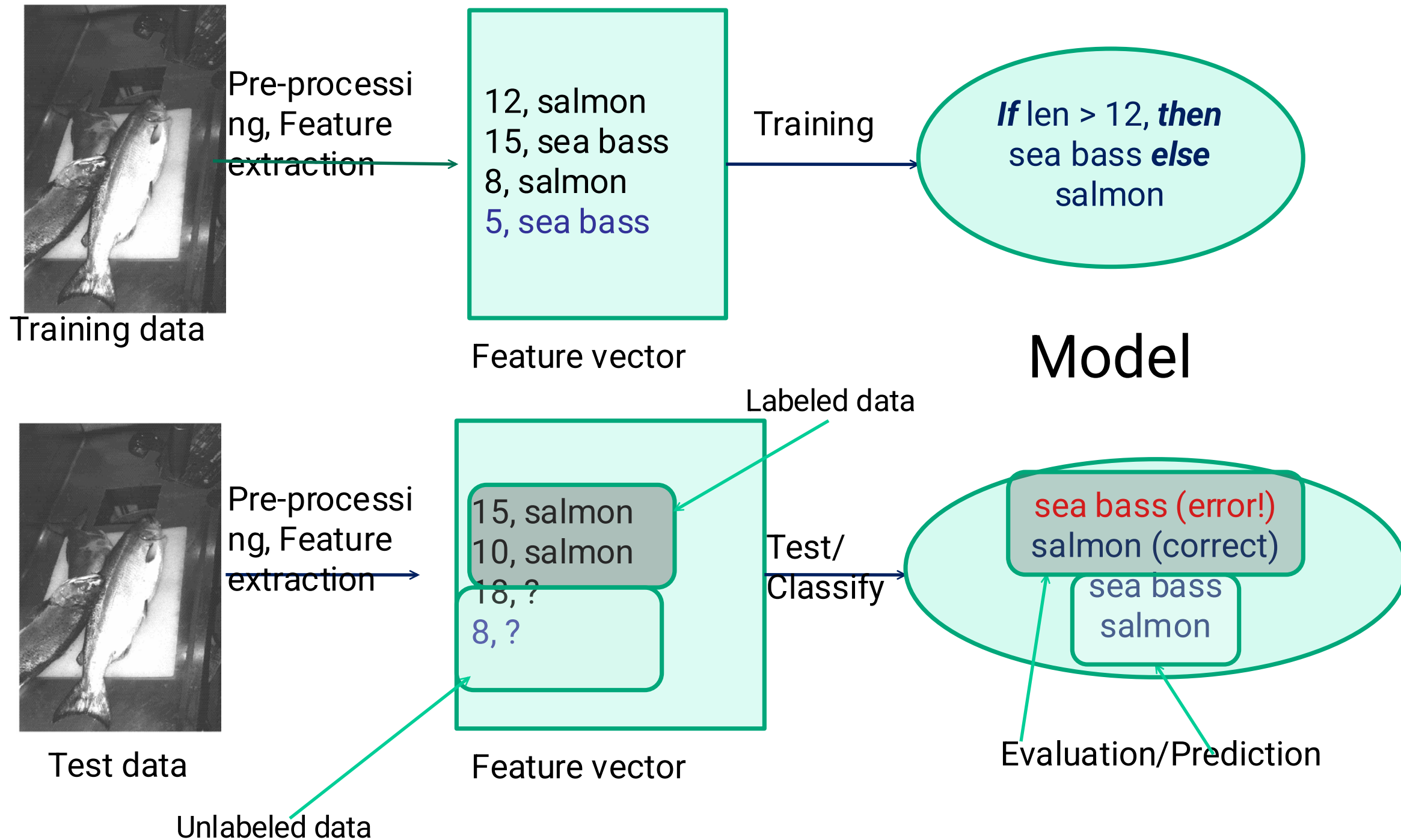
Classification

n

■ So the overall classification process goes like this →



Classification

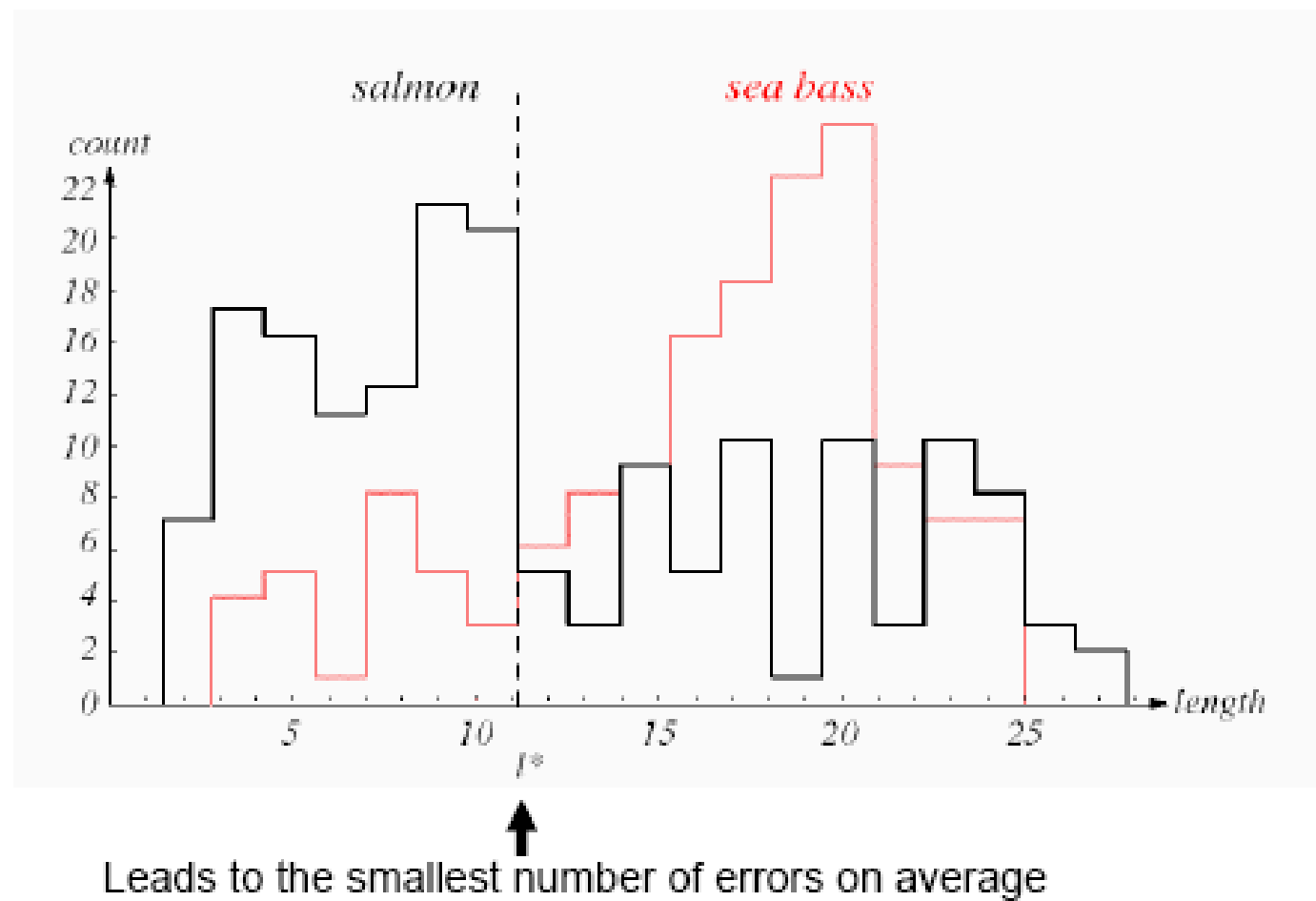


Classification

- Why error?
 - Insufficient training data
 - Too few features
 - Too many/irrelevant features
 - Overfitting / specialization

Classification

Histograms of the length feature for the two categories

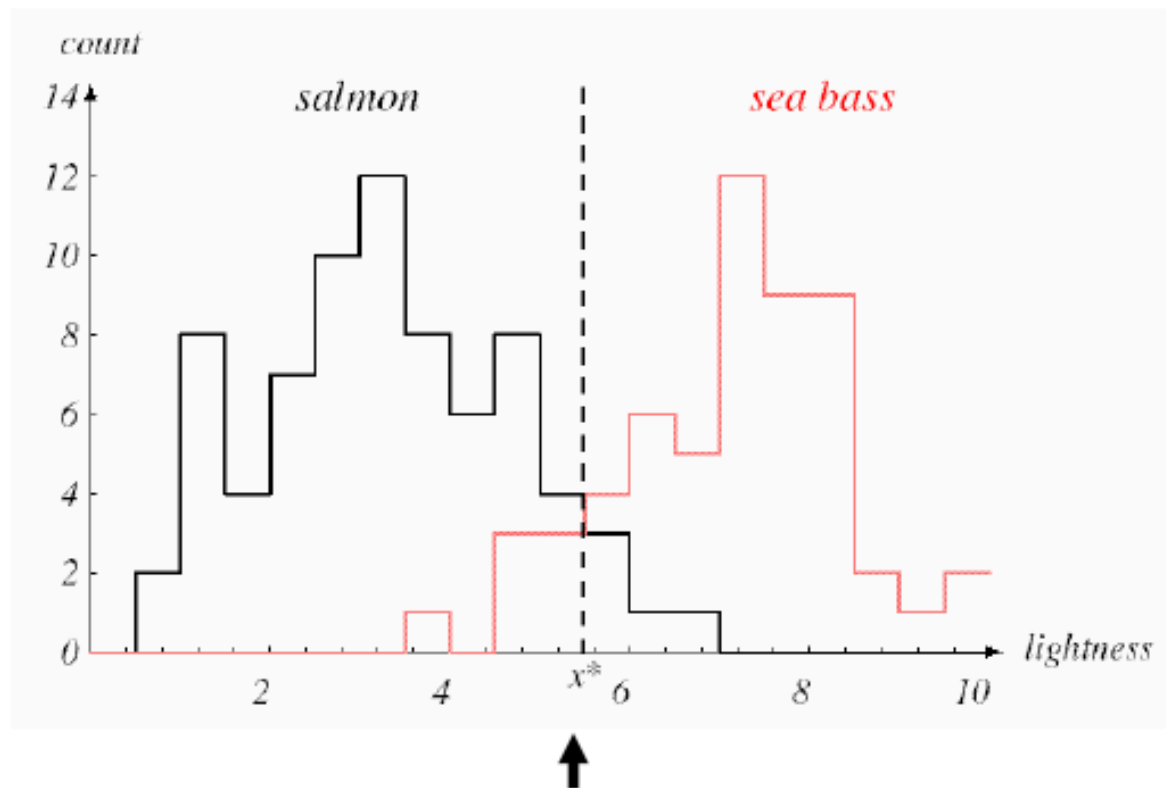


We cannot reliably separate sea bass from salmon by length alone!

Classification

- New Feature:
 - *Average lightness of the fish scales*

Histograms of the lightness feature for the two categories

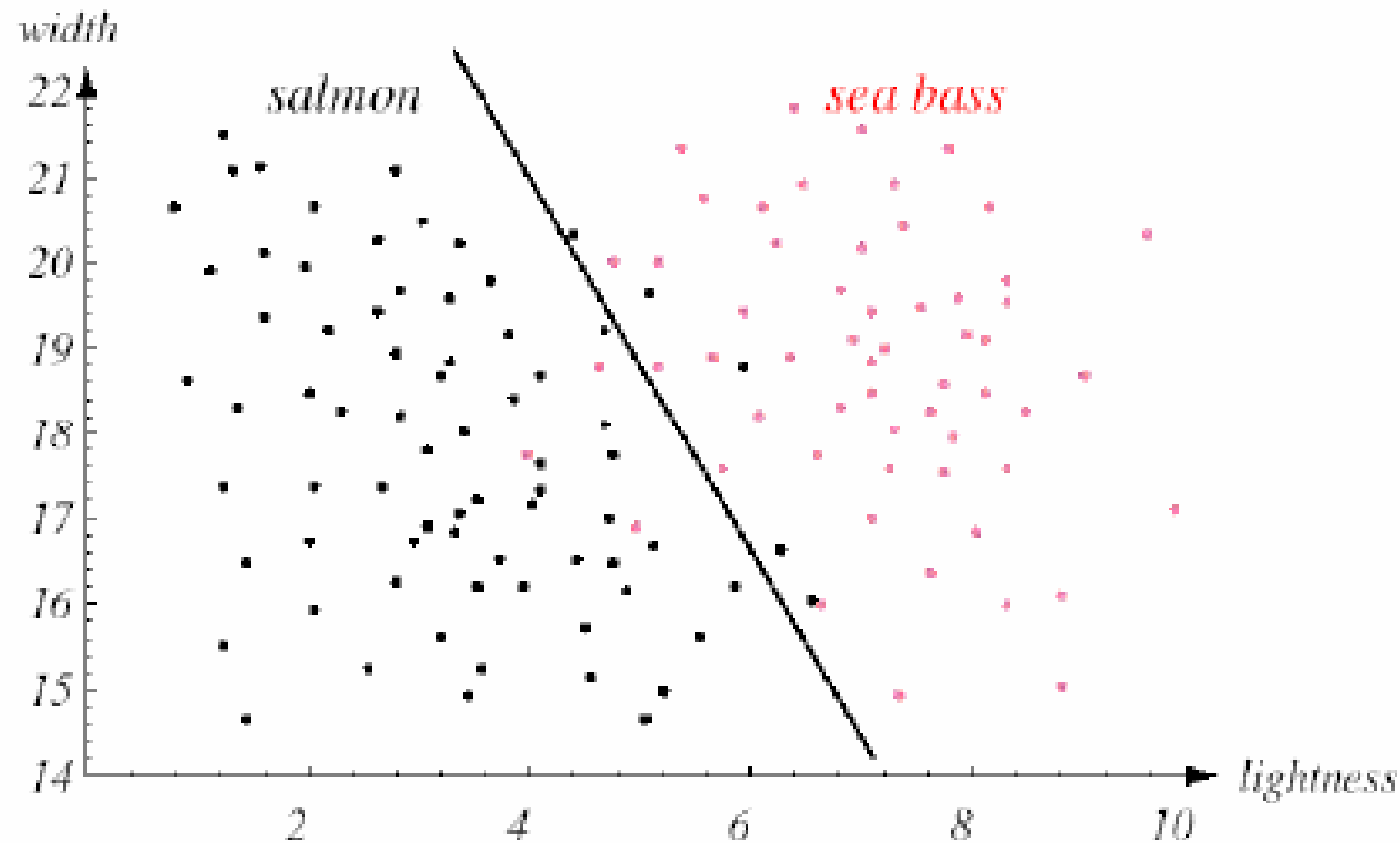


Leads to the smallest number of errors on average

The two classes are much better separated!

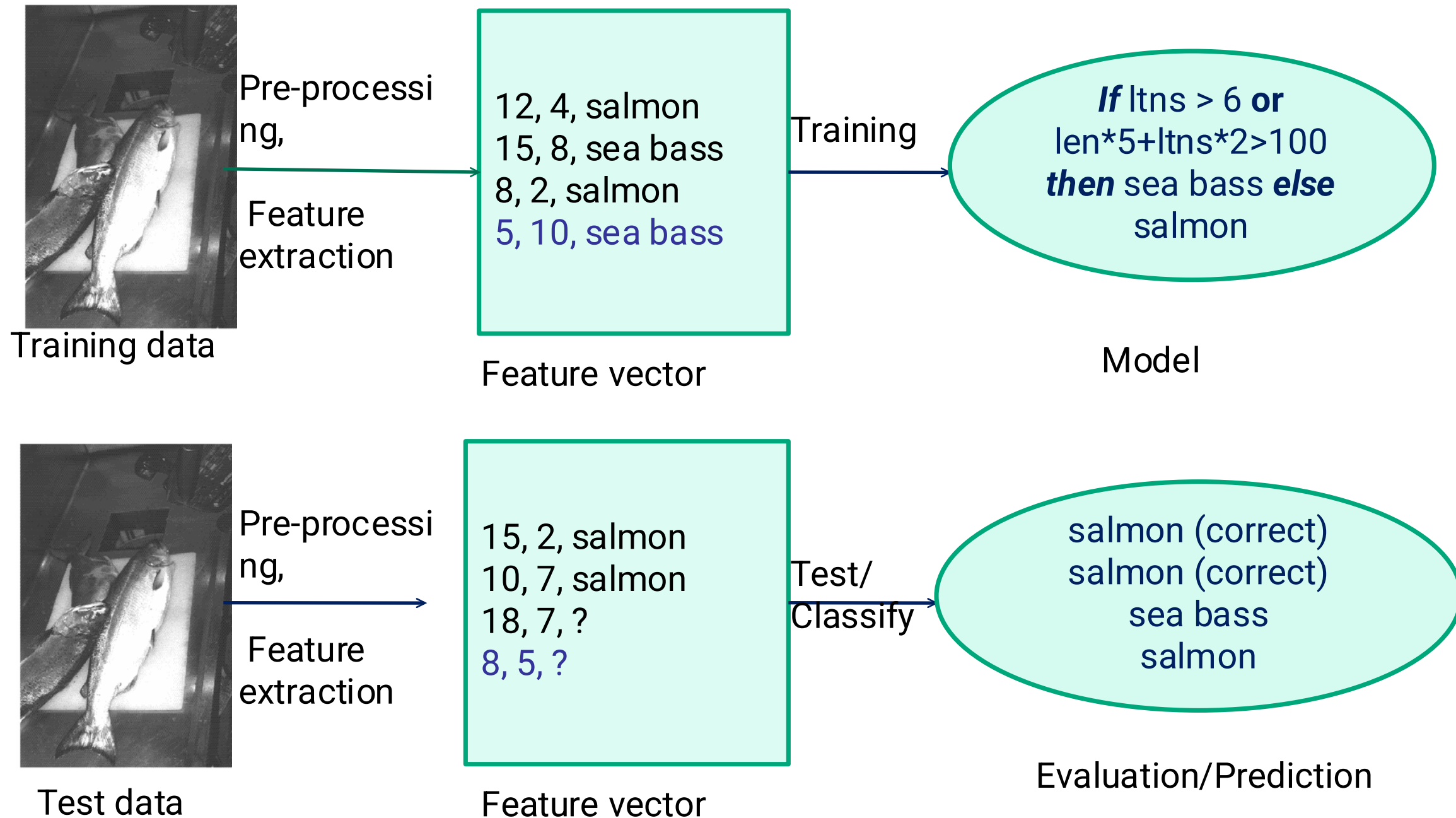
Classificatio

n



Decision rule: Classify the fish as a sea bass if its feature vector falls above the decision boundary shown, and as salmon otherwise

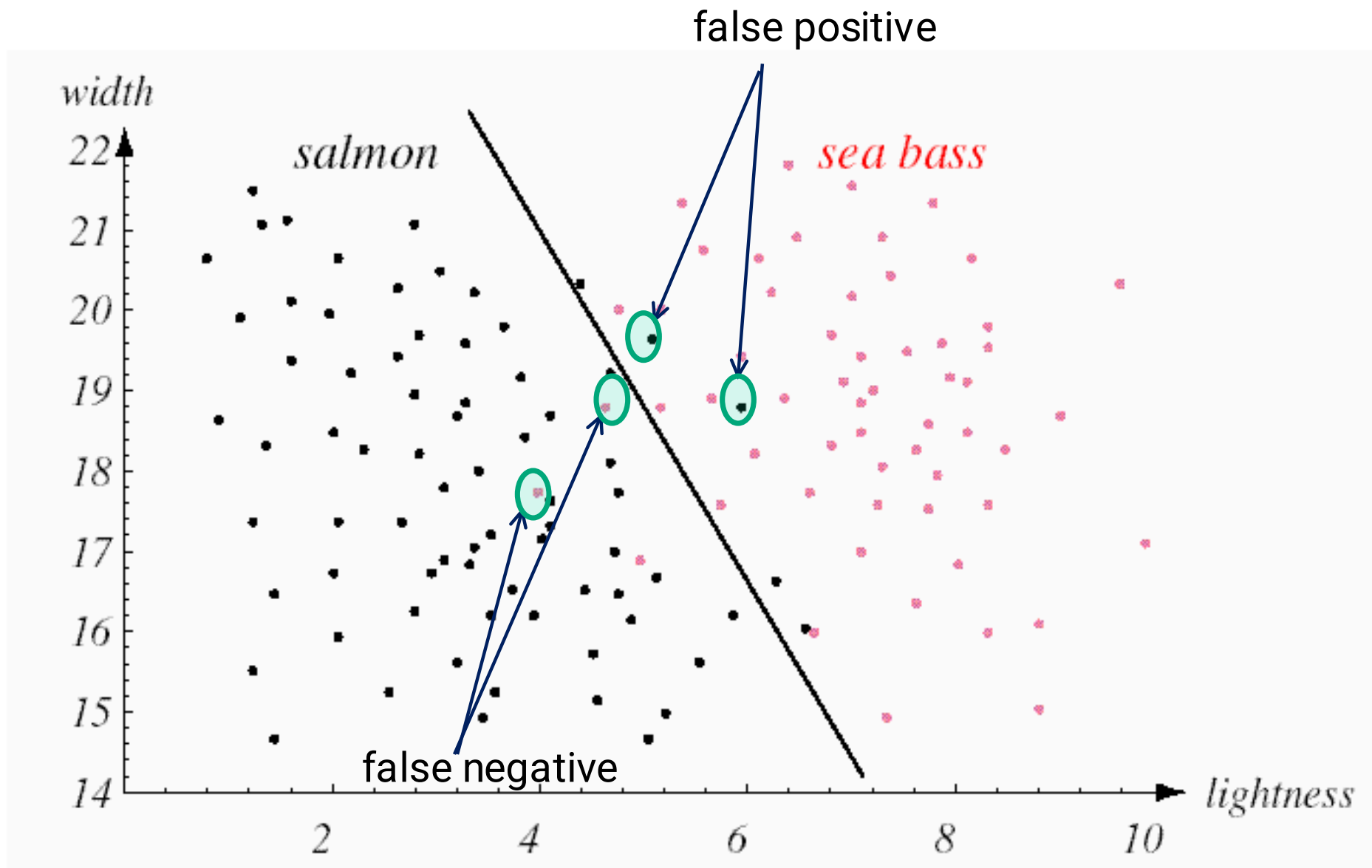
Classification



Classification: Performance

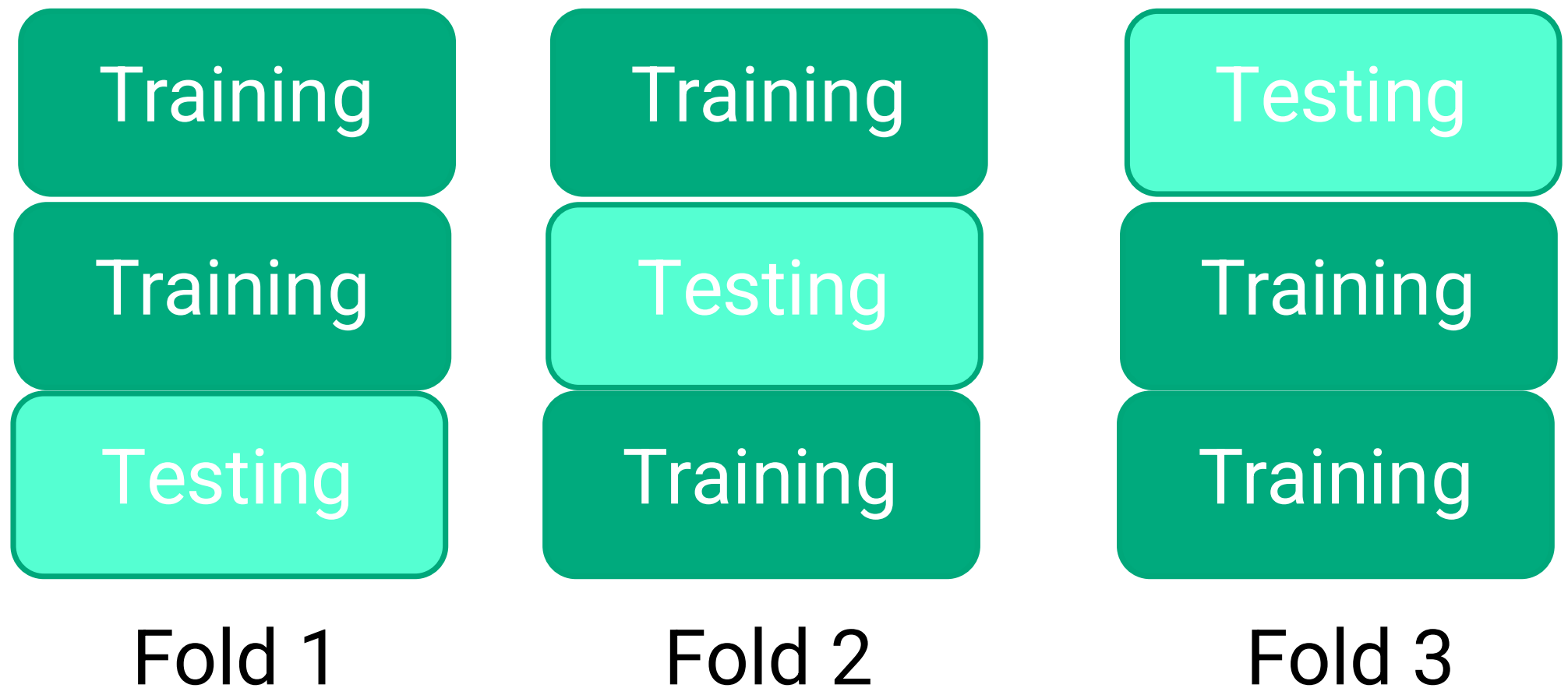
- Accuracy:
 - % of test data correctly classified
 - In our first example, accuracy was 3 out 4 = 75%
 - In our second example, accuracy was 4 out 4 = 100%
- False positive:
 - Negative class incorrectly classified as positive
 - Usually, the larger class is the negative class
 - Suppose
 - **salmon is negative class**
 - **sea bass is positive class**

Classification



Classification

- Cross validation (3 fold)





CLUSTERING

Clustering

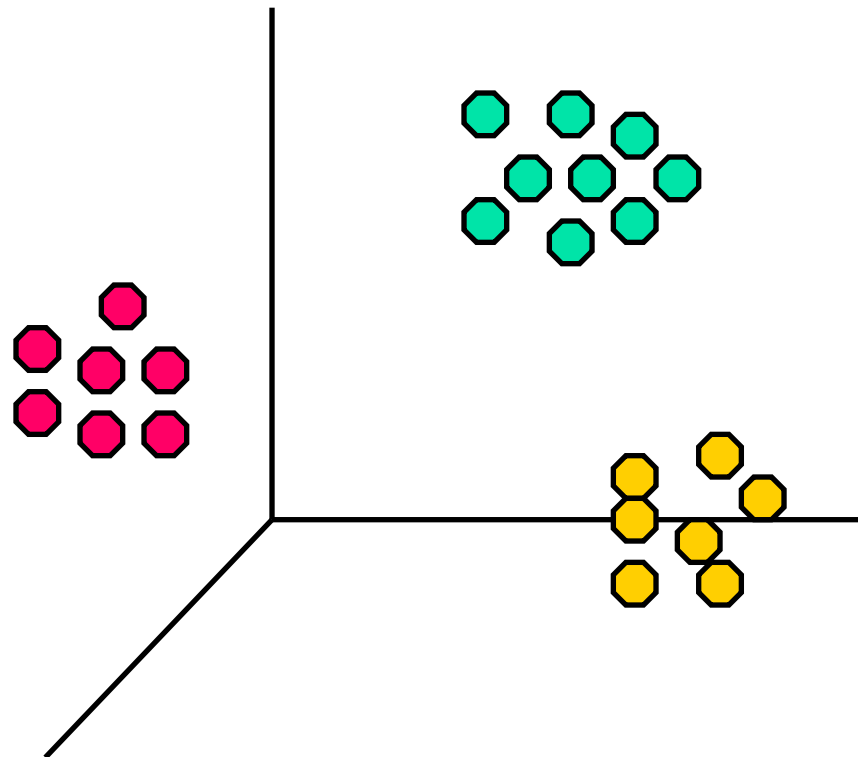
- Given a set of data points, each having a set of attributes, and a similarity measure among them, find clusters such that
 - Data points in one cluster are more similar to one another.
 - Data points in separate clusters are less similar to one another.
- Similarity Measures:
 - Euclidean Distance if attributes are continuous.
 - Other Problem-specific Measures.

Illustrating Clustering

x Euclidean Distance Based Clustering in 3-D space.

Intracuster distances
are minimized

Intercluster distances
are maximized



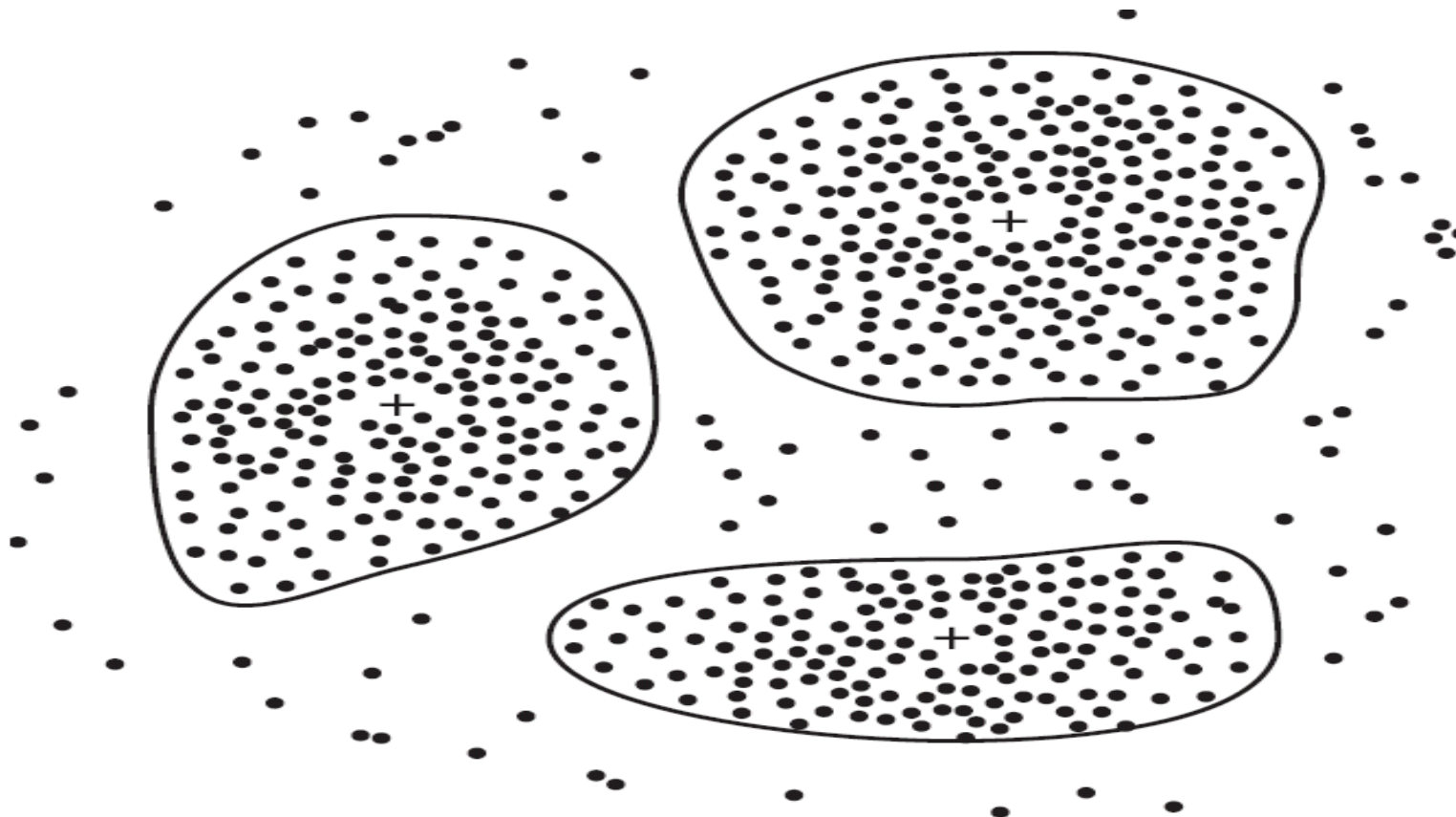
Clustering: Application 1

■ Market Segmentation:

- Goal: subdivide a market into distinct subsets of customers where any subset may conceivably be selected as a market target to be reached with a distinct marketing mix.
- Approach:
 - Collect different attributes of customers based on their geographical and lifestyle related information.
 - Find clusters of similar customers.
 - Measure the clustering quality by observing buying patterns of customers in same cluster vs. those from different clusters.

Cluster Analysis

- Class label is unknown: group data to form new classes
- Clusters of objects are formed based on the principle of *maximizing intra-class similarity & minimizing interclass similarity*
 - E.g. Identify homogeneous subpopulations of customers. These clusters may represent individual target groups for marketing.



Clustering: Application 2

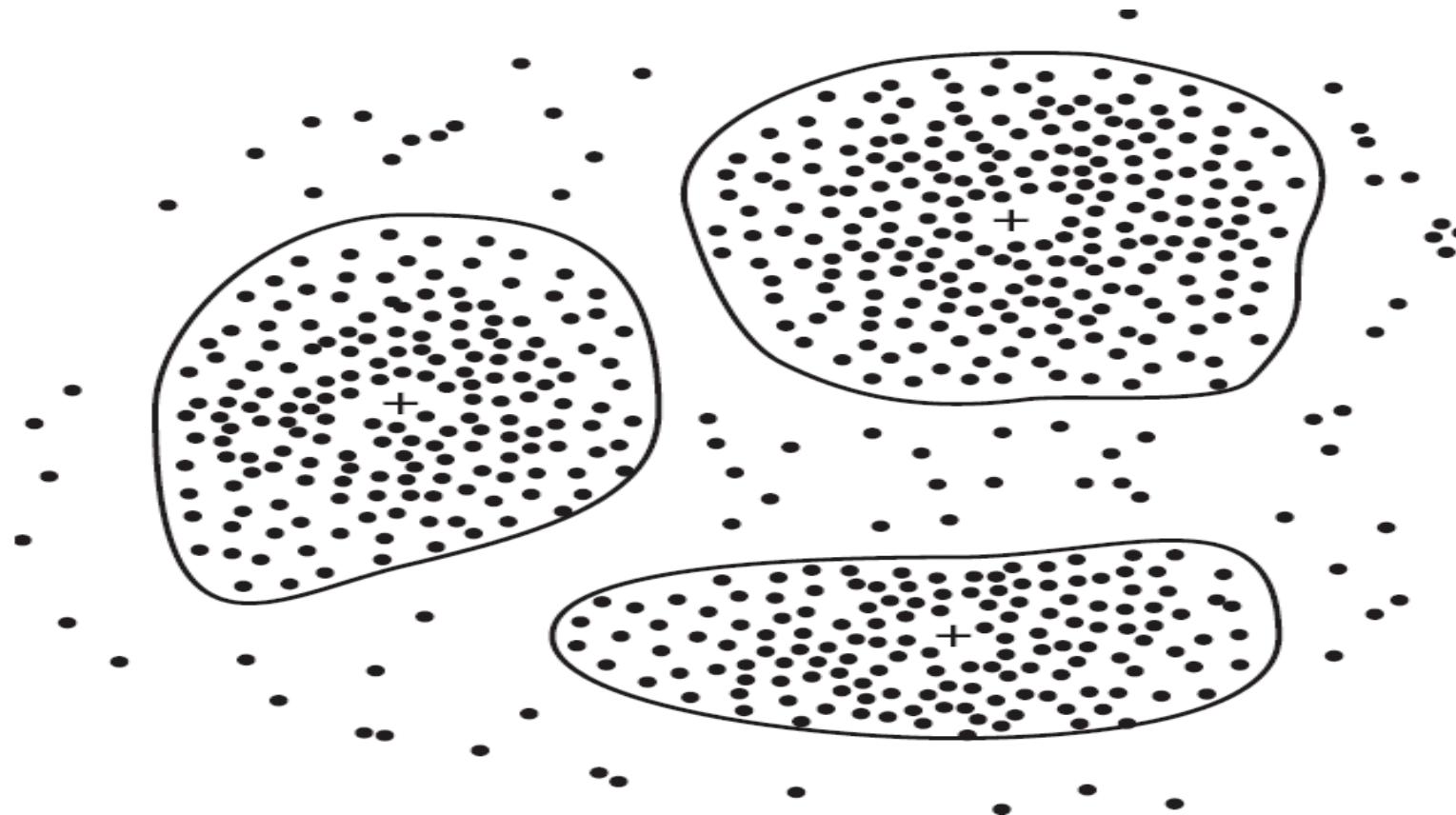
■ Document Clustering:

- Goal: To find groups of documents that are similar to each other based on the important terms appearing in them.
- Approach: To identify frequently occurring terms in each document. Form a similarity measure based on the frequencies of different terms. Use it to cluster.
- Gain: Information Retrieval can utilize the clusters to relate a new document or search term to clustered documents.

Fraud Detection

■ Outlier Analysis

- Data that do not comply with the general behavior or model.
- Outliers are usually discarded as noise or exceptions.
- Useful for fraud detection.
 - E.g. Detect purchases of extremely large amounts





ASSOCIATION RULE MINING

Association Rule Discovery

- Given a set of records each of which contain some number of items from a given collection;
 - Produce dependency rules which will predict occurrence of an item based on occurrences of other items.

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Rules Discovered:

{Milk} --> {Coke}

{Diaper, Milk} --> {Beer}



REGRESSION

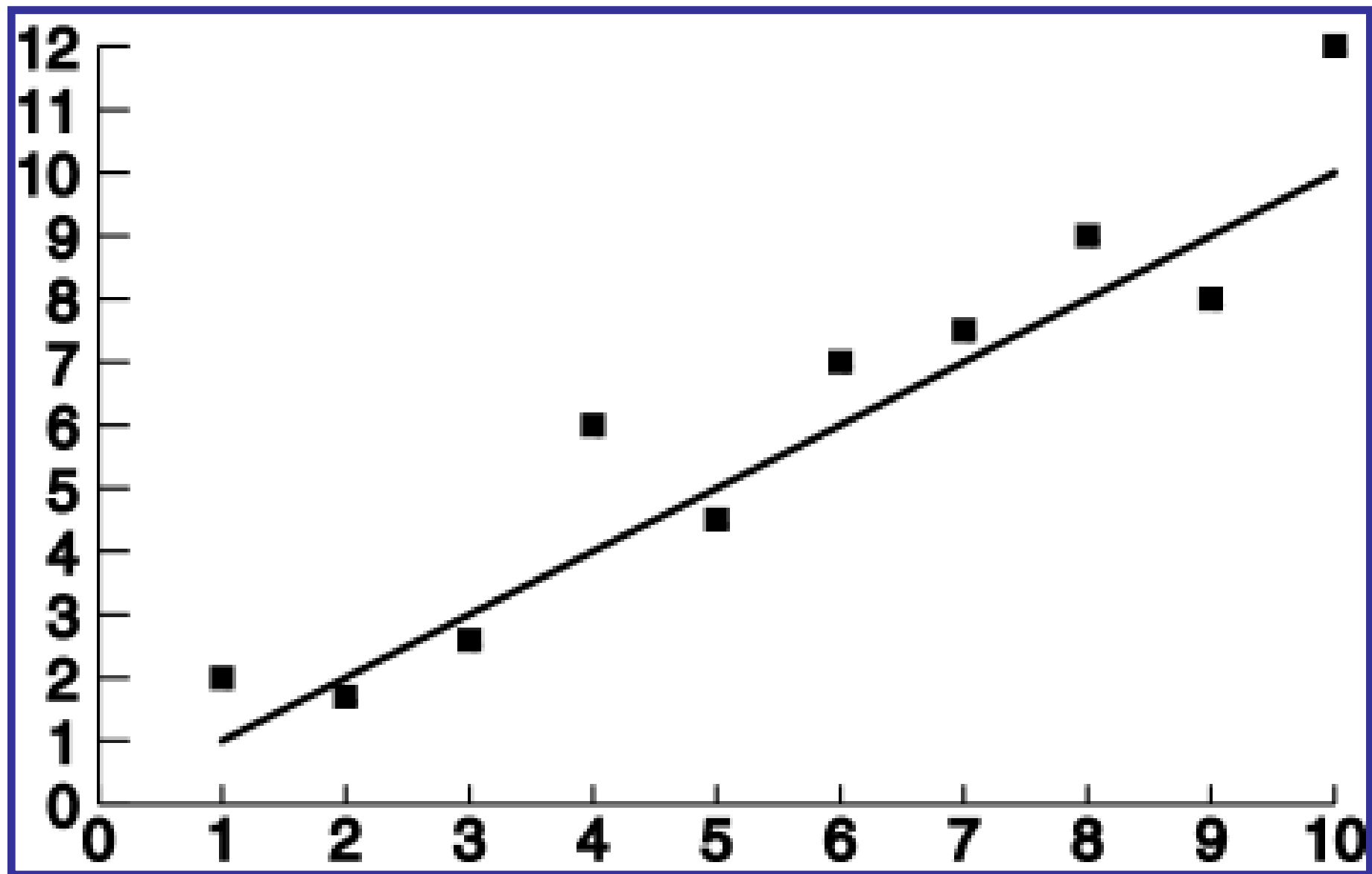
Regression

- Predict future values based on past values
- **Linear Regression** assumes linear relationship exists.

$$y = C_0 + C_1 X_1 + \dots + C_n X_n$$

- Find values to best fit the data
- Some real-world examples for regression analysis include:
 - (i) Predicting the price of a house given house features
 - (ii) Predicting the impact of SAT/GRE scores on college admissions
 - (iii) Predicting the sales based on input parameters,
 - (iv) Predicting the weather, etc.

Linear Regression



Major Issues in Data Mining



- Mining methodology and User interaction
 - Mining different kinds of knowledge
 - DM should cover a wide spectrum of data analysis and knowledge discovery tasks
 - Enable to use the database in different ways
 - Require the development of numerous data mining techniques
 - Interactive mining of knowledge at multiple levels of abstraction
 - Difficult to know exactly what will be discovered
 - Allow users to focus the search, refine data mining requests
 - Incorporation of background knowledge
 - Guide the discovery process
 - Allow discovered patterns to be expressed in concise terms and different levels of abstraction

Major Issues in Data Mining

- Data mining query languages and ad hoc data mining
 - High-level query languages need to be developed
 - Should be integrated with a DB/DW query language
- Presentation and visualization of results
 - Knowledge should be easily understood and directly usable
 - High level languages, visual representations or other expressive forms
 - Require the DM system to adopt the above techniques
- Handling noisy or incomplete data
 - Require data cleaning methods and data analysis methods that can handle noise
- Pattern evaluation – the interestingness problem
 - How to develop techniques to access the interestingness of discovered patterns, especially with subjective measures bases on user beliefs or expectations

Major Issues in Data Mining

■ Performance Issues

- Efficiency and scalability
 - Huge amount of data
 - Running time must be predictable and acceptable
- Parallel, distributed and incremental mining algorithms
 - Divide the data into partitions and processed in parallel
 - Incorporate database updates without having to mine the entire data again from scratch

■ Diversity of Database Types

- Other database that contain complex data objects, multimedia data, spatial data, etc.
- Expect to have different DM systems for different kinds of data
- Heterogeneous databases and global information systems
 - Web mining becomes a very challenging and fast-evolving field in data mining



THANK YOU